

Examining Claims of Racial Bias in the Implementation of New York City's Predictive Policing Software

Fareeza Ali, Ishika Gobin, Jackson Howes, Dana Kim
Graduate School of Arts and Sciences

Abstract

Since at least 2006, the New York Police department has maintained a database containing granular data on either all or a large portion of incidents that members of their department respond to. In 2013, a project was launched by the department shrouded in secrecy, whereby this data was leveraged by a proprietary software named Patternizr, with the goal of allocating police resources more efficiently throughout the city based on both historical and novel data, allowing for previously disparate crimes and patterns to be observed and acted upon more efficiently. The project was launched in full in December 2016, having been in development alongside other contenders for the previous three years. When the use of this system became publicly known around 2017, public outcry from both the academic space and general public expressed fears that biased data collected through years of violent and racially prejudiced policing in the department widespread throughout the early 2000s would skew this predictive system, and therefore actual police allocation, in a similarly biased direction. This paper aims to examine whether any previously existing bias in policing was exacerbated after the implementation of Patternizr. Using data from every reported police encounter in the city over 19 years for crimes involving violent felonies, we leverage machine learning and statistical analysis techniques to first determine previously existing bias through a range of demographics by police precinct, then examine changes in the rate of bias immediately after 2013, and finally monitor those rates throughout the 2010s and early 2020s as the system was implemented.

Introduction

The 2002 film *Minority Report* depicts a future Washington, D.C., in which the protagonist leads a specialized police unit that uses the foresight of three individuals known as “Precognitives” who can predict murders before they occur within a 200-mile radius. While the moral complexities raised by this story are beyond the scope of this study, its core premise closely resembles systems currently implemented by the New York Police Department (NYPD), one of many law enforcement agencies nationwide that are leveraging the rapid growth of machine learning, surveillance technologies, and access to increasingly large and granular datasets.

With the ability to know where most residents of a city are at any given time through vast arrays of cameras, license plate readers, and facial recognition software, as well as having access to a database that details calls for service received by the department and judicial records, the NYPD began the development of a customized, in-house predictive software in 2013, naming it PATTERNIZR, henceforth referred to as Patternizr. The concept is not without precedent; machine learning and predictive policing had been rolled out across the country throughout the 2000s and 2010s in a variety of purposes, including geographic inference for areas prone to future violence (Mohler, 2015; Levine, 2017), identifying patterns of behavior in individuals that may put them at future risk of perpetrating gun violence (Saunders, 2016), or even simply assisting in the recruitment and retention of officers (Lim, 2009; Weisburd, 2015; Helsby, 2018). However, prior to the development of Patternizr, there is no record of a police agency using an analogous recommendation engine to identify patterns of crimes. To borrow the description given by Patternizr’s developers, (Woods and Levine, 2019),

“The purpose of Patternizr is to take a “seed” complaint, chosen by the analyst, and compare this seed against the hundreds of thousands of recorded crimes in NYPD’s crime database. With each comparison between a crime and the seed, Patternizr generates a “similarity score” that quantifies the likelihood that the pair of crimes are in a pattern together. Once all similarity scores have been generated, Patternizr ranks the pairs in descending order by similarity score and returns a list for the analyst to review. The complaints most likely to be in a pattern with the seed complaint appear at the top of the list. After reviewing these ranked results, the analyst can decide which complaints should be grouped together into a pattern.”

In response to this implementation, a variety of social and legal scholarship, as well as media attention, decried the system as being prone to exacerbate existing biases, most commonly racial bias. Infamously,

the NYPD in the 1980s through the 2010s was notorious for intentionally overpatrolling majority black neighborhoods, stopping civilians with no reasonable suspicion, and detaining them for the purposes of a Terry stop (more commonly known as the stop and frisk). Such actions had profound effects on the neighborhoods, making them some of the most striking examples of casualties of the war on drugs.

This paper aims to examine these concerns empirically, given that the social and legal scholarship raises a valid argument that appeals to a fundamental tenet of data science: the quality of a data set and the methods by which it is collected directly impacts the accuracy of any model trained on it. We ultimately aim to determine whether the implementation of Patternizr in New York City exacerbated previously existing racial biases.

Literature Review

Owing to a lawsuit between the NYPD and Palantir regarding the collection, retention, and usage of data collected by the former but stored by the latter in and around 2017, the existence of Patternizr was revealed to the public (New York Civil Liberties Union [NYCLU], 2017; Brayne, 2020). Following this disclosure, a great deal of concern was raised by the legal and academic communities regarding both the secrecy and the very existence of such a system, which we summarize here for context (Brayne, 2017; Richardson et al., 2019).

The literature seems to indicate that the department was dramatically rehabilitated, both in their public perception and their actual policies and procedures, by the mid-2010s (Fagan et al., 2016; Floyd v. City of New York, 2013). What is of concern, both to the authors and the community at large, is whether or not the usage of data procured prior to this rehabilitation renewed biases the department had attempted to shed by unnecessarily allocating police resources to minority neighborhoods and individuals (Lum & Isaac, 2016; Brayne, 2020).

In 2023, a state superior court found that the usage of sealed or expunged arrest records by Patternizr to predict the probability of future crime to be illegal, and ordered that that method be barred from the algorithm (Matter of NYCLU v. NYPD, 2023). More generally, law reviews have taken issue with how the system skews officers' perceptions of reasonable suspicion or probable cause (Ferguson, 2017; Joh, 2019). By utilizing such a system, the responsibility of having reasonable suspicion or probable cause before detaining, searching, or investigating a citizen is at least partially removed from the officer, posing serious legal questions. Probable cause, for instance, is defined as a situation wherein a reasonable person would be assured to a high degree that a crime was likely to occur, had occurred, or was occurring (Illinois v. Gates, 1983).

It is also worth noting that 2013 was not only the year that the NYPD opted for and began the development of Patternizr, but the year that a federal judge in the class action case *Floyd v. City of New York* ruled that the practice of "stop and frisk" was an unconstitutional violation of the Fourth Amendment rights not only of the plaintiffs, but of the rights of the minority populations of New York as a whole (Floyd v. City of New York, 2012; (Floyd v. City of New York, 2013). In part, this case examined whether or not the area where Floyd experienced his stop and frisk was an area with a higher-than-normal crime rate that could lend weight to the officers' claims of reasonable suspicion, a claim that was dismissed. The court found that the department had violated civil liberties by systematically conducting stop and frisks in a racially discriminatory manner (Floyd v. City of New York, 2013).

The scientific literature regarding Patternizr is sparse, owing in large part to its secret implementation. For instance, a Freedom of Information Law request in 2019 submitted to the city requesting documentation regarding the system was rejected on the grounds that the requested documents involved "non-routine techniques and procedures" (NYCLU, 2019). As a result, the only direct insight available regarding the actual processing flow of Patternizr comes from a 2019 article written by the two designers of the system (Woods & Levine, 2019).

According to the developers, Patternizr assists individual investigators in finding patterns of crime through the use of an analogous recommendation engine, which matches descriptions of a known crime or series of crimes with descriptions of previous ones, offering a recommendation to the investigator as to where to deploy resources. The software detects pairwise similarities in 37 feature

spaces between the crime described by the investigator and the database using techniques developed by Tong et al. (2013). These pairwise similarities are then fed into a random forest model from scikit-learn that has been trained on bootstrapped random samplings of the crime type in question, and optimized for highest average precision, although no specifics are given as to how this is measured. The crimes with the highest similarities are then displayed to the investigator, along with suspects and locations (Woods & Levine, 2019). Of specific note, the authors write that they attempt to mitigate bias by only binning crimes “to very coarse squares (over nine miles on a side, for a total of roughly 11 boxes possible across the geography of New York City) to “minimize the possibility that higher-resolution geographies could be used by the model as a proxy variable for sensitive attributes (such as race)” (Woods & Levine, 2019).

Beyond that, any specifics regarding the system are piecewise intuitions for this study. We can infer from the aforementioned court finding regarding sealed arrest records that, up to and including part of 2023, Patternizr placed some weight on these records, a fact not mentioned by Woods and Levine (Matter of NYCLU v. NYPD, 2023). The NYPD uses a digital surveillance system called Domain Awareness System (DAS), which includes an array of sensors, cameras, and license plate readers throughout the city. The recent addition of Flock cameras to NYC's DAS illustrates the extent of data collected by the surveillance system, with weight being applied to everyday, non-criminal movements via license plate readers (Garvie et al., 2016). A substantial portion of the controversy surrounding the DAS is derived from the system's use of facial recognition in non-criminal applications (Garvie et al., 2016; Richardson et al., 2019).

Some ambiguity exists regarding the scale of Patternizr's implementation into the DAS for the NYPD prior to its full-scale deployment. Dates given by Holak (2019) for the implementation of Patternizr for the NYPD through DAS are as follows:

- Development of Patternizr began in and around 2013, and some level of implementation began around the same time.
- Patternizr was deployed at full production scale into the backend of the DAS in December 2016.
- In January 2017, 100 civilian analysts were hired to assist the department in crime analysis (Holak, 2019).

Study Design

A deeper understanding of the design, implementation and data sources of Patternizr provides valuable insight and guides our research question, but it should be reemphasized here that the purpose of this research is not to test or challenge the claims made by Wood and Levine regarding the efficacy of the system in its mission of improving the efficiency of police work in the most densely populated city in the country and reducing crime.

To the contrary we take them at their word; we assume that Patternizr works as advertised, was adopted uniformly across the department, and the outputs of its various models are used in a significant enough fashion to warrant their claim of a 6% reduction in crime (Woods & Levine, 2019). Instead, we intend to test the claim made in the social literature that bias present in the data has exacerbated existing bias in the city. We also take the following two uncontroversial assumptions from the social literature:

- Firstly, that some significant portion of the data used by Patternizr was procured in a period of time wherein policing in the city was decidedly biased against marginalized communities.
- Secondly, the decrease in violent crime across the city in the 1990s and 2000s was inversely related to an increase in socially responsible, equitable policing, although it was certainly not egalitarian by any means.

With these assumptions in mind, we proceeded to apply various statistical tests and a machine learning technique to investigate our research question.

Data Preparation

Data Collection

We downloaded the data directly from NYC Open Data, a government program run by the city's

Office of Technology and Innovation, aimed at increasing transparency in the city government. The data contains standardized information regarding every complaint the department responded to from 2006 through 2024. In this context, "complaint" refers to a request for service or investigation, which come in a variety of forms, such as those that are taken on the scene of a crime, over the phone, or by anonymous tip. It is updated annually to reflect that year's complaints, and is maintained regularly. Our iteration was accessed on November 15, 2025. The original, raw dataset was downloaded as a CSV file, then converted to a Data Frame containing 9,491,946 rows of complaints with 33 features each.

Initial Analysis and Cleaning

Since our analysis focuses exclusively on police complaints, we restricted the dataset to the following variables: *crime occurrence date*, *crime reported date*, *police precinct*, and *suspect race*. We kept only complaints where the crime classified as a violent felony, including but not limited to homicide, rape, and kidnapping. We added a new column labeled *time period*, to distinguish whether each complaint occurred before or after the implementation of Patternizr, with August 1st, 2013 as the cutoff date between these periods. Given the ambiguity discussed above regarding the implementation of the system, we opted to use the most conservative possible assumption as our bifurcation date. The data contained the racial groups: Black, White, Black Hispanic, White Hispanic, Asian/Pacific Islander, American Indian/Alaskan Native.

The original dataset contained additional contextual fields, such as detailed crime descriptions and victim information, however, we excluded these variables because they were not relevant to our research question and would introduce unnecessary noise into the analysis. NYC is divided into 78 police precincts, numbered between 1 and 123 and distributed across the city's 5 boroughs. For this analysis, we narrowed our focus to the top 20 precincts with the highest recorded complaint counts (Figure 1).

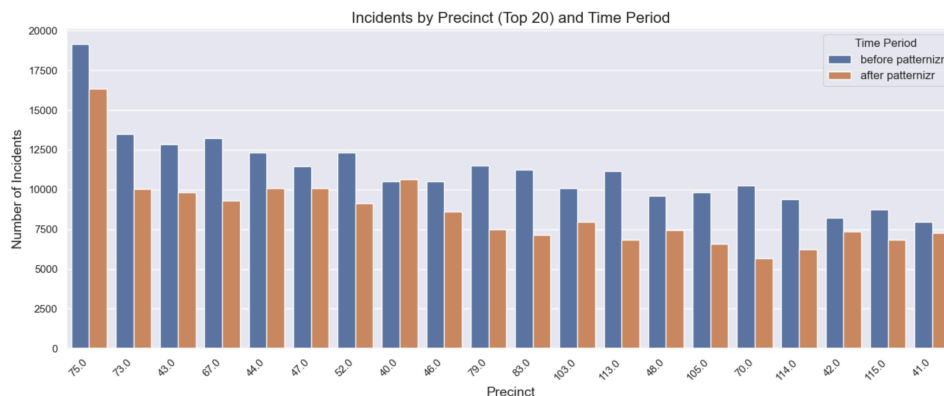


Figure 1. Top 20 precincts with most complaints before and after Patternizr

Bootstrapped Analysis of Missing or Null Values

The original raw data from the city contained approximately 537,000 rows of Null, missing, unknown, or otherwise empty rows for demographic data (age, sex, and race). In order to ensure that there was no attempt to obfuscate the demographic identities of suspects on the part of the department, willfully or otherwise, we utilized a bootstrapped difference of proportions test. Although the number of complaints with null demographic data was substantial, given the large disparity in sample sizes between the null and non-null groups (approximately 537,000 to 9 million), we opted for a bootstrapped approach. This allowed us to empirically determine the confidence intervals for the difference in proportions without relying on the theoretical assumptions of standard parametric tests.

To accomplish this, we first took our cleaned data frame and created a new column indicating whether there was missing, null, or otherwise empty data in any of the demographic columns. This data was then bifurcated into the two time periods (before or after Patternizr). We then found the observed difference in proportions of missingness between the two periods. This observed difference served as a

baseline for generating our bootstrap sample distribution, which represents how much the difference in proportions of missingness would vary due to random sampling error.

To create this distribution, we generated 5,000 bootstrap samples for each time period and calculated the difference in missingness rates between time periods for every iteration. This resulting distribution of 5,000 difference values allowed us to construct stable confidence intervals for the possible differences in missingness rates between the two periods given random resampling, which can be thought of as the true difference. In the course of this test, it was found early on that no entry contained an empty value for either the suspects or victims sex, so the demographic category of sex was excluded from further analysis.

Results of Bootstrapped Analysis of Missing or Null Values

Figure 2 shows the likely range of the true difference in proportion of missingness between the two time periods by demographic category of suspects and victims. With 95% confidence we can conclude that the true difference in proportion of missing values for the age of suspects and victims increased after Patternizr. The same can be said for the race of suspects, albeit very marginally. Only the race of victims decreased after Patternizr, again by a slim margin.

As indicated by the lack of zero in the 95% confidence intervals, we found that the changes in missingness across all categories were statistically significant (Figure 3). The lack of zero in these confidence intervals indicated that the true change in missingness was likely non-zero. However, the magnitude of these changes varied drastically. While suspect age saw a large increase in missingness (~22%), the increase in missing suspect race data was almost negligible (~1.2%). Given that our research question focuses primarily on racial bias rather than age, we determined that this marginal 1.2% shift in racial data missingness was not substantial enough to invalidate dropping rows with null demographic data before conducting our analyses. In fact, we elected to drop all rows containing null demographic data. We acknowledge that while this removes rows with missing race, it also excludes a substantial number of rows due solely to missing age. This is as a limitation to our subsequent analysis but was necessary to maintain a consistent dataset for the exploratory analysis of examining bias in age and sex.

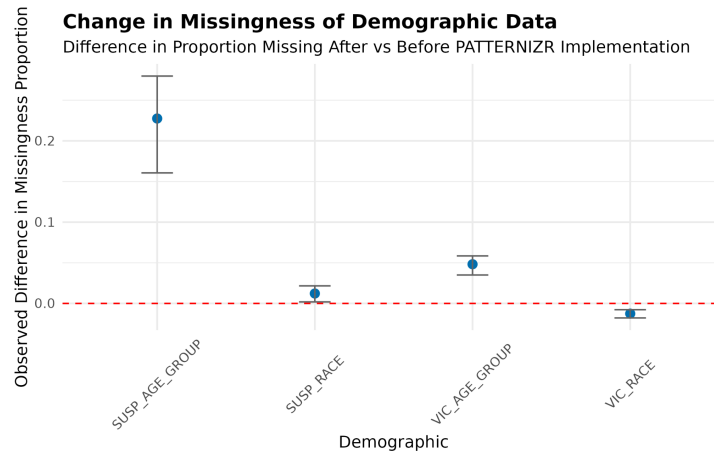


Figure 2. Bootstrapped Change in Proportion of Missingness after Patternizr

Bootstrap Results for Change in Missingness of Demographic Data						
Violent Crimes, NYPD Complaints (2006-2024)						
Demographic Attribute	Mean Before PATTERNIZR	Mean After PATTERNIZR	Observed Difference	Bootstrap Mean Difference	95% CI Lower	95% CI Upper
SUSP_AGE_GROUP	0.00666	0.23417	0.22751	0.22721	0.16062	0.27973
SUSP_RACE	0.12323	0.13545	0.01222	0.01223	0.00192	0.02163
VIC_AGE_GROUP	0.00242	0.05065	0.04822	0.04824	0.03500	0.05847
VIC_RACE	0.09920	0.08665	-0.01254	-0.01253	-0.01783	-0.00769

Figure 3. Bootstrapped Confidence Intervals for the Difference in Proportion of Missingness after Patternizr

Statistical Methods

T-test for Significant Difference in Mean Monthly Complaints per Racial Group

The first question to consider is whether the overall volume of complaints changed significantly after the implementation of Patternizr. To answer this question, we used the t-test to compare the average monthly number of complaints for each race in each precinct from before and after Patternizr to detect if there was a statistically significant difference. We accomplished this by subsetting the data from each precinct, then further subsetting each precinct-specific dataset with data from each racial group (i.e. for precinct 40 we will have 6 separate datasets, one for each racial group in that precinct). Then we created a new dataset from these subsets that contained information on the total number of complaints per month. Before proceeding with the t-test on each precinct-specific and racial group-specific dataset with total monthly complaints, we checked whether the assumptions of normality and equal variance were violated to determine whether more robust alternatives were necessary.

The results from this test allow us to distinguish between changes that are statistically meaningful or changes that simply reflect broader population growth or temporal trends. In other words, the t-test asks whether observed after Patternizr increases in complaints for each racial group in each precinct represents real shifts in policing patterns, rather than increases driven solely by demographic changes in the city.

Chi-Square Test for Difference from Expected Ratio per Racial Group

The t-test of two independent samples indicated whether, for a given racial group in a given precinct, there is a statistically significant change in volume of average monthly complaints after Patternizr. However, this test was an isolated analysis of whether each racial group in a given precinct changed after Patternizr. To get a comparative understanding, we investigated whether one racial group's change in monthly average complaints is significant in its proportion relative to other racial groups' proportion of complaints. In other words, we scrutinized whether, in a given precinct, the composition of race differs given the time period. To answer this question, we utilized the chi-square test of independence to compare the observed values of racial composition from before and after Patternizr and whether they differ significantly from if there was truly no relationship between race and time period. We examined one precinct at a time, then within each precinct-specific dataset, we split data by time period of either complaints from before or after Patternizr. Then in these two time period-specific subsets for each precinct, we created a table of counts of how many complaints there were for each racial group. With these before and after tables of racial counts, we applied the Chi-Square test to see whether the observed counts after Patternizr deviate significantly from the expected counts from before Patternizr.

After determining whether the racial composition in each precinct changed significantly after Patternizr, we examined the residuals to determine the magnitude and direction of any statistically significant changes. The residuals indicated how many standard deviations above or below the expected value a given race in a given precinct was after Patternizr. In this case, "value" refers to the ratio of that racial group relative to all other racial groups.

ANOVA for Difference of Mean Monthly Complaints by Race and Time Period

While the Chi-Square test and residuals provided insight on the change in racial composition in complaints after Patternizr for the top 20 precincts, these results did not indicate whether any specific racial group's change after Patternizr is significant relative to another specific racial group. To answer this question, we conducted a two-way ANOVA, which indicated whether there was a statistically significant difference in average monthly complaints based on time period and/or racial group.

We subsetting the data to analyze one precinct at a time, then created a new dataset that contained the total complaints in every month for each racial group. With this new dataset of monthly complaint counts by racial group and time period, we applied the standard ANOVA. Then, we used the residuals from the standard ANOVA model to check whether the assumptions of normality and equal variance between groups have been violated using Shapiro-Wilk test and Levene test. These assumptions were

violated, so we used the Aligned Rank Transformed (ART) ANOVA, the non-parametric alternative. The results from the initial ANOVA test indicated whether time period alone, race alone, or time period and race together had a statistically significant impact on the average monthly complaints in each precinct.

For statistically significant interaction effects, we continued our analysis by conducting a post-hoc test comparing each unique racial group pairing both before and after Patternizr. This determined whether, within a given precinct and time period, one racial group was over- or underrepresented in the average monthly complaints relative to the other racial group in the comparison.

Time Series Analysis of Reporting Delays and Monthly Complaint Counts

Given the pairwise differences between Black and White suspects in our analysis thus far, we further examined disparities in each racial group's interactions with the police and their communities through a time series analysis of crime reporting delays. This analysis of forecasted reporting delays may provide insight into how police or communities might respond to crimes depending on the suspect's race.

We also used time series modeling to forecast the next 36 months of complaint counts for both racial groups to examine any temporal trends in crime rates. For this portion of our analysis, we used the full dataset encompassing all NYPD precincts, and subsetting it to observations where the suspect race was identified as either "Black" or "White." Observations including Hispanic sub-descriptors were excluded.

As described in Gamage (2026), time series data is measured over equal, regular intervals of temporal ordering. Accordingly, we adopted a time-domain approach and transformed it into a time series-ready dataset by taking the monthly mean of the time variable we looked at. For reporting delays we calculated the number of days between when each crime was said to have occurred and when that crime was reported to the police. We then aggregated those delays by months and found the mean days of delay for each month-year from January 2006 to December 2019. We repeated this monthly aggregate and mean process for our time series model on forecasting the monthly number of crime complaints.

We utilized the AutoRegressive Integrated Moving Average (ARIMA) model for both time series analyses, as described by Gamage (2026), which allows us to capture temporal patterns in the data and generate reliable forecasts. Gamage (2026) emphasizes that the ARIMA model requires data to be stationary given that it is a type of linear regression that performs most effectively when predictors are uncorrelated and independent, and when statistical properties are constant over time. Using the Augmented Dickey Fuller (ADF) test, we evaluated stationarity and applied differencing as necessary, until ACF plots revealed little to no persisting autocorrelation. We selected the autoregressive (p) and moving average (q) orders by examining ACF and PACF plots and minimizing Akaike Information Criterion (AIC), which is an estimation of prediction error.

To ensure we used the best ARIMA model parameters for each time series, we relied on comparing a combination of the Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and ACF1 correlation scores of our models to those of mean, naive, and drift forecasts.

Results

T-test for Significant Difference in Mean Monthly Complaints per Racial Group

Most of the datasets that were subset by precinct and racial group did not meet the assumptions of normality or equal variance. However, there were enough observations ($n > 30$) to overcome this lack of normality. To account for the lack of equal variance in many of the datasets, we applied Welch's t-test to all the precinct and race-specific datasets for simplicity and robustness.

Across the top 20 precincts, every precinct had at least one racial group with a statistically significant change in mean monthly complaints after Patternizr (Figure 4). However, not all racial groups experienced significant changes in every precinct, suggesting that the impact of Patternizr on monthly complaints may vary by both race and location. White Hispanic and Black suspects exhibited statistically significant changes in monthly complaints most frequently relative to other racial groups after Patternizr. In fact, they were found to have changed significantly after Patternizr in all 20 precincts. This also means that White Hispanic and Black suspects were found to have changed significantly across the widest

geographic range among racial groups.

Following closely, Black Hispanic and White suspects were found to have changed significantly after Patternizr in most but not all of the precincts analyzed, implying that in some locations these suspects were not impacted by Patternizr. The racial group with the least frequent significant changes in monthly complaints was Asian/Pacific Islander, which showed highly significant but geographically concentrated increases, particularly in Queens. While these results do not prove causation, they show that the timing and location of racial shifts align closely with the deployment of predictive policing.

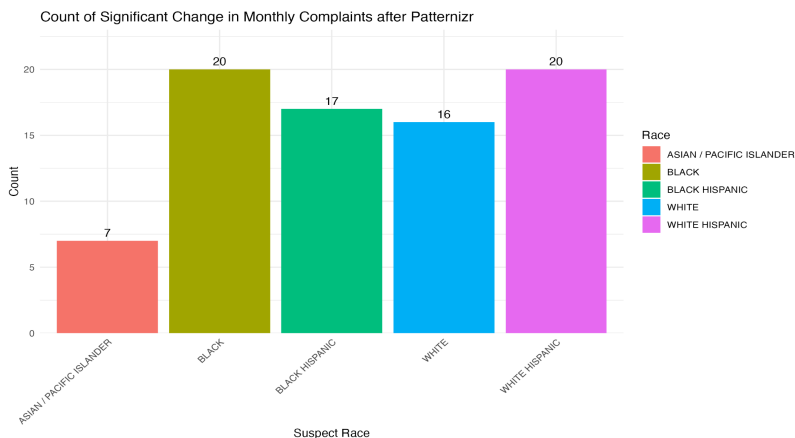


Figure 4. Count of Statistically Significant Changes in Monthly Complaints per Racial Groups Across Top 20 Precincts

Chi-Square Test for Difference from Expected Ratio per Racial Group

In all but 2 of the 20 precincts included in our analysis, we observed a statistically significant change in racial composition after the implementation of Patternizr. Accordingly, for these 18 precincts, we conclude that the monthly number of complaints involving suspects of a given racial group varies by time period. This suggests that the implementation of Patternizr was associated with changes in the racial composition of suspects involved in complaints, altering which racial groups experienced more interactions with the police after its introduction. However, these results alone do not indicate which specific racial groups were affected by Patternizr, nor how they were impacted (including whether the impact was a relative decrease or increase in representation in complaints).

Closer inspection of the residuals in chi-square test results revealed that among precincts with statistically significant changes in racial composition after Patternizr, the shifts most often reflected underrepresentation of historically marginalized racial groups and overrepresentation of historically privileged racial groups. For example, in precinct 40, Black suspects became underrepresented and White suspects became overrepresented after Patternizr relative to their proportions prior to implementation. When aggregating results across the 18 precincts, we find that White and Asian/Pacific Islander suspects are more frequently overrepresented than underrepresented, whereas Black suspects are more frequently underrepresented than overrepresented. Figure 5 illustrates the number of precincts in which each racial group was over or under represented relative to its pre-Patternizer proportions.

Although these results might suggest that Patternizr “corrected” biases associated with the overpolicing of historically marginalized groups, such as Black suspects, it is important to recall that the Chi Square test indicates whether the composition of a variable of interest changed significantly relative to the expected composition. Importantly, the expected values are derived from the dataset, which are unlikely to be devoid of racially biased overpolicing. Although the observed ratio of historically marginalized racial groups is lower than predicted from the expected distribution after Patternizr, it is very likely that the expected ratio after Patternizr is based on an already biased overrepresentation of Black suspects before Patternizr. Given overwhelming evidence on the marginalization of Black communities, while Black suspects tend to become more underrepresented after Patternizr (which can be interpreted as

“less bias”), this is relative to likely already racially biased policing before Patternizr.

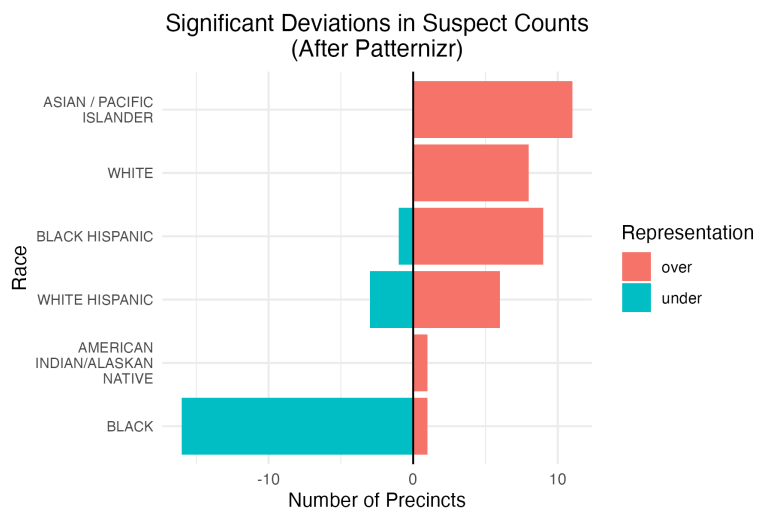


Figure 5. Count and Direction of Statistically Significant Racial Group Proportions After Patternizr across 20 Precincts

It is worth noting that in the Chi Square, if one group becomes “underrepresented”, the other groups must mathematically be “overrepresented” since we are looking at overall proportions. Thus, while it also seems that White and Asian suspects were overrepresented, given evidence that White and Asian suspects tend to be underpoliced, this result is more likely a mathematical byproduct of the underrepresentation of Black suspects relative to the likely biased proportions of Black suspects before Patternizr. We should be careful not to assume that these results mean that White or Asian suspects were suddenly overpoliced after Patternizr. What we can conclude is that Black suspects became underrepresented relative to their likely biased overrepresentations before Patternizr.

ANOVA for Difference of Means from Two Independent Variables

Across all top 20 precincts, both the main effects and the interaction effects were statistically significant. This indicates that for these precincts, the combined effect of time period and racial group influence the average monthly number of total complaints per precinct. Although, the results also indicate that time period and racial group alone affect the average monthly number of total complaints per precinct, since the interaction effect was significant for all precincts, we cannot accurately describe the impact of Patternizr on the monthly average number of complaints without also specifying which racial group we’re considering.

These results alone don’t tell us which pairs of racial groups across which time periods were disproportionately affected. To gain this insight, we must specifically test each unique combination of racial group pairs in both time periods, which clarifies which pairs of racial groups are disproportionately found in the complaint records both before and after Patternizr. This allows us to compare these disproportionate impacts before and after Patternizr.

ANOVA: Frequency of Disparity

Across the 20 precincts analyzed, there were 20 unique racial groups pairings that were statistically significant either before or after Patternizr (Figure 6). The statistically significant racial pairings that occurred most frequently were Black–White, followed by Black–Asian/Pacific Islander and American Indian/Alaskan Native–White. This means that when statistically significant differences in average monthly complaints were observed across precincts, whether before or after Patternizr, they most often occurred between these racial group pairings.

The results also indicated that statistically significant differences in average monthly complaints

occurred more frequently before Patternizr than after its introduction. Despite the reduction in significant differences post-Patternizr, these findings should not be interpreted as evidence that Patternizr “corrected” bias, as the pre-Patternizr data itself is almost certainly racially biased.

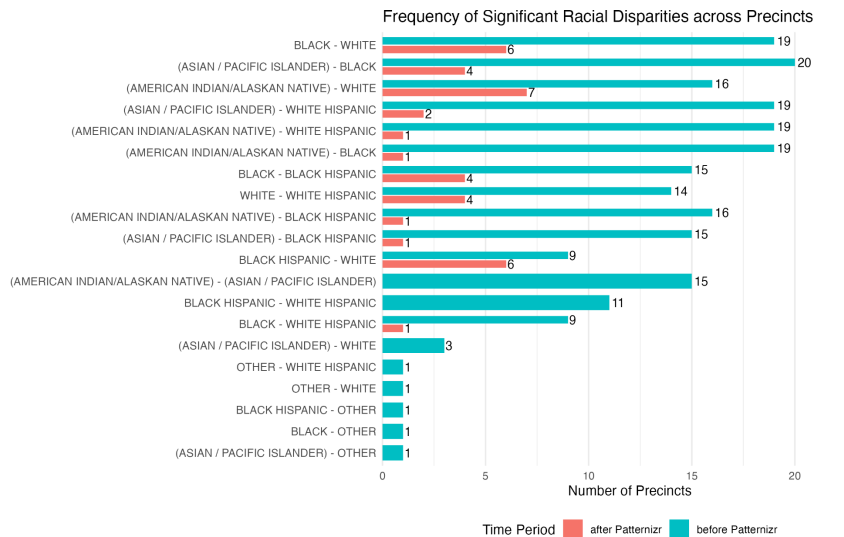


Figure 6. Count of Racial Pairings with Statistically Significant Disparities in Average Monthly Complaints across 20 Precincts

ANOVA: Direction of Disparity

As seen in Figure 7, among the top 3 racial pairings that most frequently exhibited significant disparities in average monthly complaints, we found varying difference in representation after Patternizr for most racial groups when compared specifically to another group.

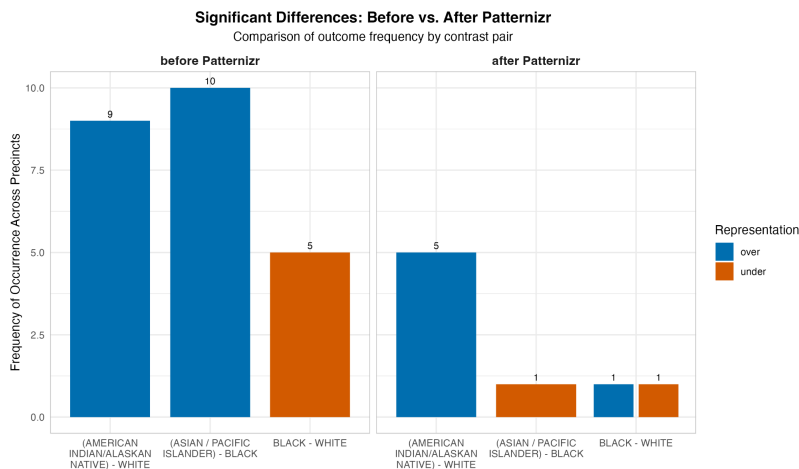


Figure 7. Count and Direction of Statistically Significant Racial Pairings for Top 3 Most Commonly Found Pairings

American Indian/Alaskan Native suspects were more frequently represented in complaints than White suspects across precincts, both before and after the implementation of Patternizr. This indicates that American Indian/Alaskan Native suspects were disproportionately overrepresented in complaints compared to White suspects across the entire time range. Although the gap between these two racial groups decreased after Patternizr, American Indian/Alaskan Native suspects remained overrepresented.

Asian/Pacific Islander suspects are found more frequently in the monthly average complaints across precincts than Black suspects before Patternizr. But this changes after Patternizr, where

Asian/Pacific Islander suspects are found less frequently in the complaints than Black suspects. This means that before Patternizr, Asian/Pacific islander suspects are overrepresented in complaints compared to Black suspects, but afterwards Black suspects become overrepresented compared to Asian/Pacific Islander suspects. This implies that after the implementation of Patternizr, there were disproportionate complaints of Black suspects compared to Asian/Pacific Islander suspects.

Black suspects are found less frequently than White suspects before Patternizr, but afterwards there is one occurrence of Black suspects being overrepresented and one occurrence of Black suspects being underrepresented compared to White suspects. This result tells us that there is a relative increase in Black suspects in the complaints after Patternizr compared to White suspects.

These results in combination fit our understanding of the historical marginalization of Black and American Indian communities, the model minority myth of Asian American communities, and the historical privilege of White communities. While it's not accurate to say that White or Asian people do not commit as much crime as Black or American Indian suspects, we can infer from these results that they are not showing up in police records as often as Black suspects, which could be a reflection of bias against BIPOC (Black and Indigenous People of Color) communities.

Time Series Analysis of Reporting Delays

Reporting Delay of Crimes Involving Black Suspects

The original time series of average monthly reporting delays for complaints involving Black suspects was confirmed as stationary, so no differencing was required. We compared non-seasonal models across p (0–3), d (0–1), and q (0–2) using AIC, BIC, and AICc. The non-seasonal ARIMA(2,0,1) model outperformed the seasonal alternative suggested by auto.arima, so we utilized ARIMA(2,0,1) for our analysis.

The selected model outperformed mean, naive, and drift benchmarks, resulting in an RMSE of 7.47 and almost zero autocorrelation ($ACF1 = 0.009$), compared to the mean forecast ($RMSE = 7.81$, $ACF1 = 0.213$) and naive forecast ($RMSE = 9.76$, $ACF1 = -0.478$). The relatively small improvement over benchmarks reflects a volatility in reporting delays, potentially because our dataset does not differentiate between types of crimes (and some crimes are more likely to be reported immediately than others). The model's 36-month forecasts that reporting delays will stabilize around 15 days, with wide confidence intervals that reflect the series' high variability.

Reporting Delay of Crimes Involving White Suspects

Again, the original time series of average monthly reporting delays for complaints involving White suspects was also confirmed as stationary. We also compared non-seasonal models across the same p , d , q parameters as above with Black suspects. The ARIMA(1,0,1) model had the lowest AIC, though it only slightly outperformed the seasonal alternative.

The selected model outperformed naive and mean benchmarks, achieving an RMSE of 26.33 and almost no autocorrelation ($ACF1 = -0.071$), compared to the naive ($RMSE = 36.75$, $ACF1 = -0.550$) and mean ($RMSE = 26.92$, $ACF1 = 0.063$) forecasts. Note that we did not use MAPE as the primary evaluation metric for this series due to 17 months having reporting delays under 5 days, which inflate percentage-based error measures. The high variability in reporting delays reflects the variance in this data as well. The 36-month forecast projects reporting delays stabilizing around 35–40 days, with wide confidence intervals also reflecting the series' high volatility.

A Comparison of Black vs White Reporting Delays

Comparing the reporting delays of crimes involving Black suspects to that of crimes involving White suspects reveals two things. The first is that there is no observable pattern in either time series that indicates a meaningful shift after Patternizr was implemented. Second, crimes involving White suspects are forecasted to have 3x more days in reporting delays than crimes involving Black suspects. This could be a reflection of a social phenomena, where people are less likely to call the police or report a crime if

they don't believe that it will yield any results, and more likely to report a crime if they believe that police will be responsive. It could also just be a reflection of different types of crimes and how they are reported, assuming the specific types of violent felonies where suspects are White are different from those where suspects are Black.

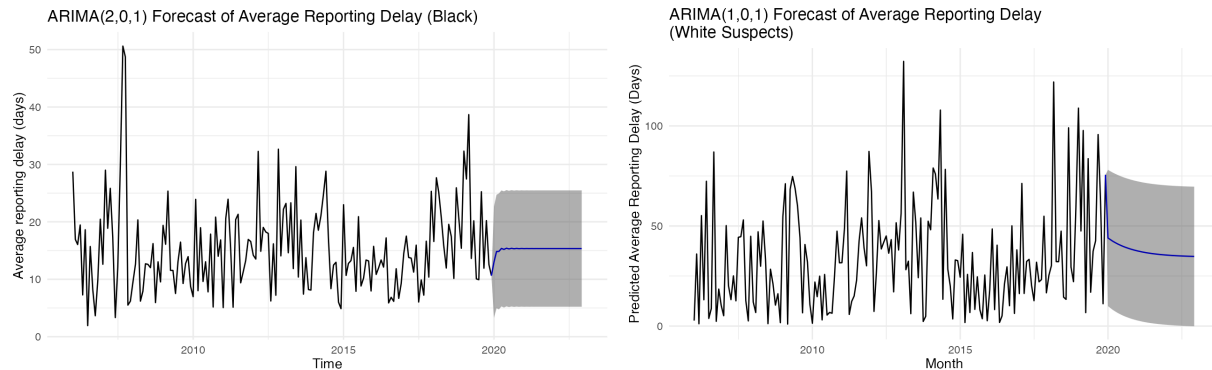


Figure 8: Best fit ARIMA time-series model forecasts of crime report delays of incidents with Black suspects vs. White suspects.

Time Series Analysis of Monthly Complaints

Complaints Involving Black Suspects

The original time series was non-stationary, so we used first-order differencing, which produced a significant ADF result. We also visually inspected the ACF/PACF plots to confirm adequate stationarity. We then compared non-seasonal models across p (0–3), d (0–1), and q (0–2), however, auto.arima identified a seasonal ARIMA(1,0,0)(2,1,2)[12] with substantially lower AIC. As a result, we selected the seasonal model.

The seasonal model substantially outperformed mean, naive, and drift benchmarks, achieving a MAPE of 6.12% and near-zero residual autocorrelation ($ACF1 = 0.067$), compared to the naive (MAPE = 9.93%, $ACF1 = -0.101$) and drift (MAPE = 9.98%, $ACF1 = -0.101$) forecasts. We found the resulting 36-month forecast of crime complaints involving Black suspects to be around 1,500–1,800 complaints per month, with continued seasonal fluctuations.

Complaints Involving White Suspects

We used the same process on time series data of crimes involving White suspects. Again, the data was non-stationary and we achieved stationarity through first-order differencing. The auto.arima again identified a seasonal model so we used the same p , d , q parameters on non-seasonal models and compared it to the auto.arima recommended SARIMA(0,1,1)(0,0,2)[12]. The seasonal model resulted in a lower AIC.

The seasonal model here also outperformed benchmarks, resulting in a MAPE of 12.40% and essentially zero autocorrelation ($ACF1 = 0.0005$), compared to the naive forecast (MAPE = 14.56%, $ACF1 = -0.367$) and drift forecast (MAPE = 14.61%, $ACF1 = -0.367$). The 36-month forecast projects monthly complaint counts involving White suspects to hover between 240–250 per month, again, with seasonal fluctuations.

A Comparison of Black vs White Monthly Count Forecasts

These time series forecasts show a clear increase in the average monthly crime complaints involving White suspects between 2014 and 2015, which is an increase of approximately 100 per month that never drops below 150 per month again. We see a similar jump during that time period for crime complaints of Black suspects, except that time series shows an increase of approximately 700 per month and does not drop back below 1200, aside from a dip in February 2015. These observations suggest a correlation between the time period of Patternizr's widespread adoption and a nearly doubled increase in

policing involving both White and Black suspects.

However, while the proportion of the increase is relatively equitable, the number of crimes with Black suspects are forecasted to hover around 5-7x times monthly compared to crimes with White suspects. As a result, this data shows that even post Patternizr, NYPD's policing disproportionately reflects crimes involving Black suspects, which would increase the likelihood of predictive policing outcomes that disproportionately affect the Black population.

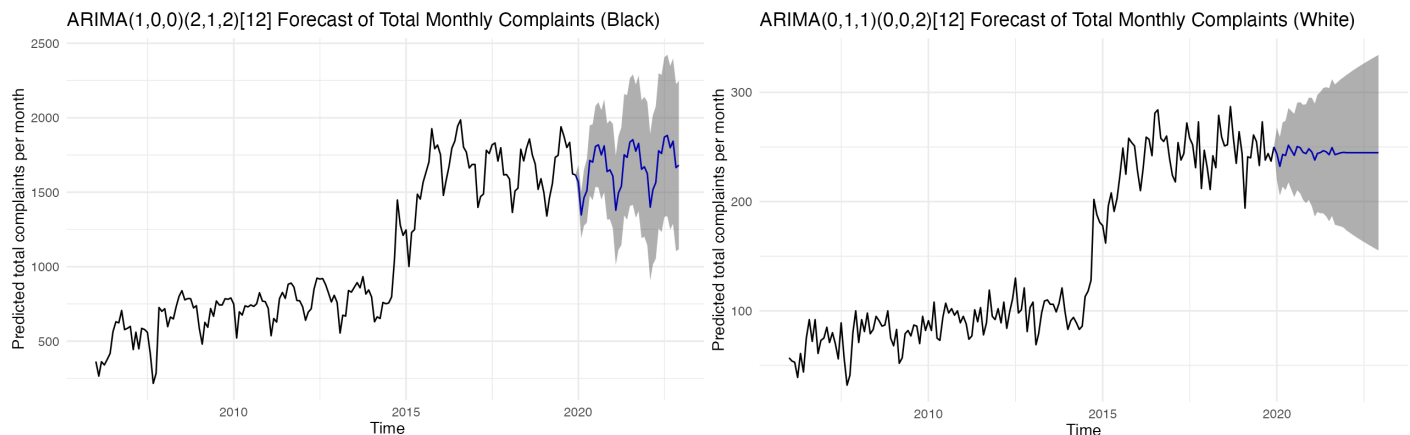


Figure 9: Best fit ARIMA time-series model forecasts of crime complaints involving Black suspects vs. White suspects

Conclusion

Over the course of this project, we set out to examine whether the NYPD's introduction of Patternizr, a predictive policing software, exacerbated racial bias in policing. Our results indicated that crimes involving Black suspects had the most consistent increase in complaints across precincts, compared to other racial groups. Further, the racial composition of violent felony crime complaints shifted significantly in 18 of the 20 precincts in the city with the highest crime counts. Although Black suspects became less proportionally overrepresented after Patternizr, this is relative to likely already extreme overrepresentation from before Patternizr. Crime complaints involving Black suspects were also forecasted to occur at least 5-7 times more frequently than White suspects. Finally, crimes were reported about 3 times faster for Black suspects compared to White suspects.

What does this mean for the residents of NYC? For Black NYC residents wondering whether Patternizr has been good news for them, the answer is: probably not. Even with Patternizr's implementation and the end of explicit racial profiling policies such as stop-and-frisk, Black individuals remain disproportionately likely to appear in crime complaints compared to White residents. The premise of predictive policing is that it is efficient and fair to allocate resources where they are likely to be needed most, but the gap in reporting delays suggests that not every suspect is addressed by law enforcement in the same way. Our statistical findings are not just numbers, each data point represents real interactions with police, real stress and real consequences for communities that have long been, and continue to be, overpoliced.

At this point, we understand that algorithms are only as good as the data that they are trained on. Once again, we highlight that Patternizr draws from historical NYPD data that has been shaped by years of policies like stop-and-frisk, policies that have disproportionately targeted Black and minority communities. While we did not conclusively find that Patternizr's implementation is associated with exacerbated racial bias, we did find that it is associated with persistent bias. The difference is, with an algorithm, real people who do harm via predatory policing are less likely to be held accountable because the program diffuses responsibility. Our results raise the question of whether predictive policing in the U.S. can ever be racially neutral in the current era, given the extensive racial profiling that has taken place across every metropolitan city in the country over the past several decades.

Our study has limitations that leave important questions unanswered. First, we did not examine downstream outcomes like arrests or convictions resulting from the recorded crimes in our dataset. Thus, we are unable to assess Patternizr’s broader impacts on justice outcomes. Second, we did not disaggregate our data or results by type of violent felony crime. Different racial groups may be associated with different crime types and some of our findings may therefore reflect crime-type distributions rather than racial bias. Third, the scope of our study does not allow us to explain why reporting delays differ between crimes involving Black suspects vs. White suspects. These differences may reflect variations in community trust in police, historical patterns in NYPD responsiveness or underlying crime-type differences. Future research would benefit from distinguishing among felony crime types to assess whether our findings persist at that level.

We provide the following takeaways from our findings: for policymakers, we underscore the need for greater scrutiny of the data used to train Patternizr and whether audits for racial bias are being conducted in an effective way. For the NYPD, any efficiency gains from predictive policing are insufficient if they come at the cost of perpetuating inequality or compromising the safety and security of certain communities in favor of others. For the public, the central questions are not whether a predictive policing algorithm ‘works’, but rather whom it works for and whether truly equitable predictive policing is possible in the United States. We leave readers with a final thought: Technology is not neutral. While predictive policing may modernize law enforcement, without confronting the historical biases embedded in its data, it risks accelerating inequality under the guise of efficiency.

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